

Some project guidelines and advice on presentation technique...



Study in *Information and Networks* – NOT programming

- Use whatever calls and routines are available to complete the task efficiently
- If you've written 100s of lines of code, you've almost certainly done something wrong
- You are not showing how clever you are if you implement some function from first principles when there's a one line call that would do it anyway!
 - Think about whether a future employer would be impressed

This is an open-ended task

- We have not “taught” you the theory behind your project. You are expected to research it.
- There is no “correct” answer
- No well-defined point of completion
- Some people “run with it” and benefit from exploring where it can take them
- Some people do the absolute minimum to satisfy the description in the notes
 - (BIG mistake!)

Report

- Two pages (meaning two SIDES) MAXIMUM, including diagrams, etc. (but not code)
- Put references on a third page if you have to
- Marks will be deducted if over-length
- So think CAREFULLY how to get the important messages across succinctly
 - This is a really important skill
- Only one student of each pair needs to upload a copy

Presentation

- During the scheduled lab time in week 10 or 11
- (Most likely) live in half the lab space.
- BOTH people need to speak, each for about half the time
- PLAN it! You have 5 minutes...
 - Who is your audience?
 - What do they already know?
 - What was the point of the exercise?
 - A practical investigation of an aspect of telecommunications BEYOND THE LECTURES AND LABS
 - So...
 - Don't waste our time telling us what we already know
 - Educate us! Tell us what you learned!
 - And make sure you present some results!
 - How many slides can we realistically get through?
 - What is the key information we should include?
 - Should we have some “back up” slides?
- PRACTICE it!

Presenting

- Speak CLEARLY
- Speak LOUDLY
 - If you think you're speaking too loudly, you're probably about right
- Be aware of the space, the most distant audience members
 - Speak to the audience; not the screen
- Don't read
 - If unsure or nervous, take some written prompts, but try to get away with just the key points
- You don't need to cram as much information as possible
 - You just need the RIGHT information
 - Slow is OK
 - Pauses can be powerful
- Try a little animation
- Practice
- And remember the purpose is to investigate and report on telecommunications – not programming – so DON'T show us lines of code!

I had a clever way of doing my transmission simulation:

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% AWGN CHANNEL %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
for snr_iter = 1:length(SNR)
```

```
    EbNo = SNR(snr_iter);
```

```
    % Calculate the SNR for an oversampled QPSK signal.
```

```
    snr = EbNo + 10*log10(k) - 10*log10(Samples_per_symbol); % calculate the SNR per sample
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% RECEIVER %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
% Add AWGN
```

```
rxSig = awgn(txSig, snr, 'measured');
```

```
rxSigFilt = upfirdn(rxSig, FiltCoeff, 1, Samples_per_symbol); % Apply the RRC receive filter and downsample to symbol rate
```

```
rxSigFilt = rxSigFilt(FilterSpan+1:end-FilterSpan);
```

```
rxSigFilt_beforesample = upfirdn(rxSigFilt, FiltCoeff);
```

```
dataOut = qamdemod((1/scale)*rxSigFilt, M);
```

```
Errors = sum(dataOut ~= data); % calculate symbol errors
```

```
BER_simulation(snr_iter) = Errors/(FrmLen*k); % calculate BER
```

```
end
```

Visuals / Slides

- Don't fill them with text (and don't overfill them with anything)
- Don't just read the slides
- Graphics, and short text points, generally work well
- Graphs have axes. They need labels!
- Will those labels be visible?
- Numbers – how many significant figures?
- When to use line graphs, points, etc.
- Error/range bars

Comparison of Analogue and Digital Transmission

- The cost advantages of digital transmission over analogue transmission become apparent when transmitting over a long distance.
- Consider, for example, a system that involves transmission over a pair of copper wires.
- As the length of the pair of wires increases, the signal at the output is attenuated and the original shape of the signal is increasingly distorted.
- In addition, interference from extraneous sources, such as radiation from radio signals, car ignitions and power lines, as well as noise inherent in electronic systems result in the addition of random noise to the transmitted signal.
- To transmit over long distances, it is necessary to introduce repeaters periodically to compensate for the attenuation and distortion of the signal. Such signal reconditioning is fundamentally different for analogue and digital transmission. With analogue transmission, all we can do is amplify the signal (and the noise!). With digital, we regenerate the digital bits and retransmit them on the next leg of the transmission.

Analogue and Digital Transmission

- Over long distances, a transmitted signal is
 - Attenuated
 - Distorted
- Additional signals also injected:
 - External interference
 - Random noise

Signal loss -> Repeaters

- Repeater function:
 - analogue: amplify the signal (and noise!)
 - digital: regenerate and retransmit

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